A nilpotent cellular automaton that remembers under noise

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Abstract

Does every positive zero-range perturbation of a nilpotent cellular automaton have a unique attracting stationary law? We construct a counterexample on $\mathbb{Z}^3$, answering a question of Marcovici, Sablik and Taati. Each site carries a bit and a bounded countdown. The deterministic rule reaches the all-zero configuration after a fixed number of steps, while independent uniform replacements renew the countdowns and allow a horizontal bit plane to retain its initial value. With stationary countdowns, the plane is exactly a north-east-centre majority process with independent gates and independent replacement faults. We calculate the gate law and apply Toom's stability theorem in the quantitative version of Gács to obtain two translation-invariant stationary laws. For the same fixed automaton, a dependency-path estimate proves uniform exponential ergodicity at sufficiently small and sufficiently large noise rates. We also construct, for every prescribed separation, a nilpotent automaton, a set of replacement events at pairwise distance exceeding that separation, and two initial configurations whose bits differ on a whole plane at all times under those common events.

AI systems assisted with this work; see "How this work was done."

1 Introduction

A nilpotent cellular automaton erases every initial configuration in a bounded time. It is natural to ask whether adding independent noise can only reinforce this loss of memory. Marcovici, Sablik and Taati asked whether every perturbation of a nilpotent cellular automaton by positive zero-range noise is ergodic [2, Problem 6.5]. We give a negative answer using a bounded countdown and Toom's majority rule. Our example is three-dimensional, and its alphabet has $2(H+1)$ symbols; the independence of the gates uses the third coordinate. We do not address dimensions one and two.

The distinction between a fixed small noise rate and a noise rate tending to zero is essential. Every fixed nilpotent rule remains ergodic under sufficiently small perturbations [2, Theorem 3.9]. In our example, noise must arrive often enough to renew a countdown before it expires. A long countdown makes this renewal effective at a noise rate still small enough for majority correction to preserve a bit. At still smaller rates the countdowns expire and erase the bits.

Here is the resulting behavior for one fixed rule.

Theorem A. There is a finite-alphabet, radius-one cellular automaton on $\mathbb{Z}^3$ that is nilpotent and whose independent uniform replacement perturbation has at least two translation-invariant stationary laws at a positive noise rate. The same rule is uniformly exponentially ergodic at all sufficiently small and all sufficiently large positive noise rates.

The explicit rule is given in Section 2; Theorems 4.2 and 5.1 prove the assertions. A purely deterministic statement isolates the renewal mechanism.

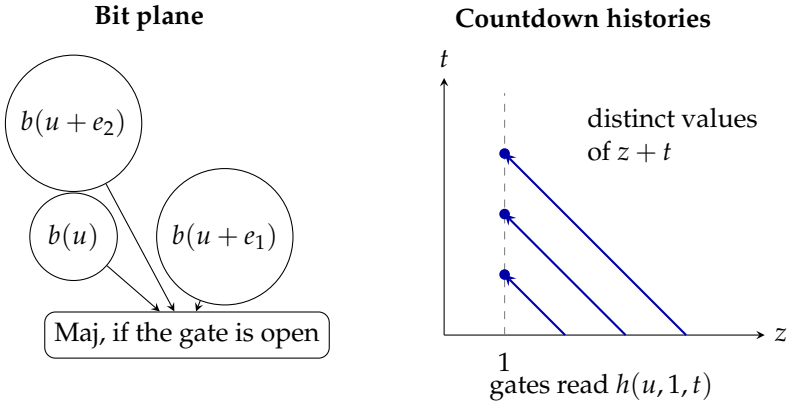


Figure 1: The bits use majority within one horizontal plane. The countdown read by each successive gate follows a different diagonal in the (z,t) coordinates. Independent randomness on these disjoint diagonals gives independent gates.

Result	Behavior under noise	Deterministic rule
Marcovici–Sablik–Taati [2]	Ergodicity under all sufficiently small perturbations of each fixed nilpotent rule.	Nilpotent
Marsan–Sablik–Törmä [4]	One-dimensional example with ergodicity at low and high noise, and non-ergodicity between.	Cyclic clock
This paper	Three-dimensional example with the same three regimes under uniform replacement noise.	Nilpotent

Table 1: The nilpotent question and the earlier two-transition phenomenon.

Theorem B. For every $d \geq 0$, there is a nilpotent cellular automaton on $\mathbb{Z}^3$ and a set of space–time replacement events at pairwise ℓ^∞ distance greater than d such that two trajectories, subject to the same replacement symbols, disagree on an entire plane at every time.

This statement is proved in Theorem 6.1. The automaton may depend on d.

Toom proved stability of north-east-centre majority under small errors [6]. We use the explicit infinite-plane estimate in Gács's exposition [1]. The extra spatial coordinate in our construction lets the countdowns move toward a fixed data plane without reusing their random histories there; Figure 1 illustrates this independence.

Marsan, Sablik and Törmä already established the phenomenon of ergodicity at low and high noise, with non-ergodicity in between, for a one-dimensional cellular automaton [4]. Their rule retains a cyclic clock in the absence of noise. Our contribution is a nilpotent rule with this behavior, and an exact reduction to a familiar two-dimensional majority process. The dependence of stationary laws on vanishing noise for nilpotent automata is also studied by Marsan and Sablik [3].

Reading guide. Sections 2 and 3 define the model and derive its independent gates. Section 4 imports the majority-stability estimate and proves coexistence of stationary laws. Section 5 gives the coupling proof for the two ergodic regimes. Section 6 proves the separated-event statement directly. The only non-elementary input is Toom's stability theorem in the stated quantitative form.

2 The rule and the noise

Write $\mathbb{N} = \{0, 1, \dots\}$. Fix an integer $H \geq 1$ and write a site of $\mathbb{Z}^3$ as (u, z) with $u \in \mathbb{Z}^2$ and $z \in \mathbb{Z}$. Its state belongs to

$$\mathcal{A}_H = \{0, 1\} \times \{0, 1, \dots, H\}.$$

The two coordinates of a configuration are denoted by b and h. Let e_1, e_2 be the positive coordinate vectors in $\mathbb{Z}^2$, and let $a_+ = \max(a, 0)$. Define the cellular automaton F_H by

$$(F_H h)(u, z) = (h(u, z+1) - 1)_+, \tag{1}$$

$$(F_H b)(u, z) = \mathbf{1}_{\{h(u,z+1)>0\}} \operatorname{Maj}\big(b(u, z), b(u + e_1, z), b(u + e_2, z)\big). \tag{2}$$

The notation $F_H h, F_H b$ refers to the coordinates of $F_H(b, h)$. Its neighborhood consists of 0, $(e_1, 0)$, $(e_2, 0)$ and $e_3 = (0, 0, 1)$. Thus its radius in the ℓ^∞ metric is one.

Proposition 2.1. *For every configuration, $F_H^{H+1}(b, h)$ is the uniform state $(0, 0)$. The least such common exponent is $H + 1$.*

Proof. Induction in (1) gives

$$h_t(u, z) = (h_0(u, z + t) - t)_+.$$

Consequently all countdowns are zero at time H. The next bit update is zero at every site, proving the first assertion. Starting with $b_0 \equiv 1$ and $h_0 \equiv H$ gives $h_t \equiv H - t$ and $b_t \equiv 1$ for $0 \leq t \leq H$. This configuration has not reached $(0, 0)$ at time H, proving minimality. □

For $0 < \varepsilon \leq 1$, the noisy rule $P_{H,\varepsilon}$ first applies F_H and then, independently at each site and time, replaces the whole symbol with probability ε by a uniform element of $\mathcal{A}_H$. More explicitly, at times $t \geq 1$ let

$$R_t(u, z) \sim \text{Bernoulli}(\varepsilon), \qquad V_t(u, z) \sim \text{Bernoulli}(1/2), \qquad U_t(u, z) \sim \text{Unif}\{0, \dots, H\}$$

be mutually independent over all indices. If $R_t = 1$, set $(b_t, h_t) = (V_t, U_t)$ at that site; otherwise use (1)–(2) from time $t - 1$. Every one-site transition probability is at least $\varepsilon/[2(H + 1)]$. The noise is therefore positive and has zero range in the terminology of [2].

Let $\Omega_H = \mathcal{A}_H^{\mathbb{Z}^3}$ with the product topology. A stationary law is a probability measure μ satisfying $\mu P_{H,\varepsilon} = \mu$. We call the PCA *ergodic* if it has a unique stationary law and the law from every initial configuration converges weakly to it. It is *uniformly exponentially ergodic* if, for some $C, c > 0$ and every finite $W \subset \mathbb{Z}^3$,

$$\sup_{x \in \Omega_H} d_{\mathrm{TV}}\big((\delta_x P_{H,\varepsilon}^t)_W, \mu_W\big) \leq C|W|e^{-ct}. \tag{3}$$

All constants here may depend on H and ε.

The space Ω_H is compact and metrizable. Since a local observable after one update depends continuously on finitely many input symbols, the transition operator is Feller. In particular a stationary law exists: any weak limit point of the Cesàro averages $T^{-1} \sum_{t=0}^{T-1} \nu P_{H,\varepsilon}^t$ is stationary. Indeed, applying $P_{H,\varepsilon}$ to this average changes its integral against a bounded continuous observable by at most twice the supremum norm divided by T. We will use this elementary compactness argument for both existence and coexistence.

3 Countdowns and independent gates

Write $r = 1 - \varepsilon$ in this section. At a site (u, z) and time t, the countdown reads its past along

$$(u, z + a, t - a), \qquad a = 0, 1, \ldots, t. \tag{4}$$

Along this diagonal the value of $z + t$ is constant. If the most recent replacement on it occurred a steps ago, the present countdown is $(U - a)_+$. In the absence of such a replacement it is $(h_0(u, z + t) - t)_+$. Only the last H update positions can affect the present value: a reset of age at least H has already reached zero, as has every possible initial countdown after H steps.

Lemma 3.1. *The countdown process has a stationary product law $\pi_{H,\varepsilon}^{\otimes \mathbb{Z}^3}$. Its one-site marginal is the law of $(U - A)_+$, where U is uniform on $\{0, \ldots, H\}$ and A is independent with*

$$\mathbb{P}(A = a) = \varepsilon r^a, \qquad a \in \mathbb{N}.$$

For every $t \geq H$ the countdown configuration has this product law from every initial configuration, and it is independent of the initial countdowns.

Proof. For one diagonal, the geometric variable A records the age of the last replacement in an indefinitely extended past. At one further step its age is zero with probability ε, and otherwise increases by one, which preserves this geometric law. The reset mark U is independent and uniform, so the stated marginal is invariant under the countdown transition.

At a fixed time, distinct sites follow disjoint diagonals in (4). Product initial countdowns and independent replacement marks therefore give independent countdowns again. This proves stationarity of the product law. At time $t \geq H$, truncating each diagonal to ages $0, \ldots, H - 1$ gives exactly the same value: if there has been no replacement in these H positions, the value is zero. This is the same function of the same finite set of marks for every initial countdown configuration. Disjointness of the diagonals proves the final assertions. □

Fix a horizontal bit plane, say $z = 0$, and define its gate by

$$G_t(u) = \mathbf{1}_{\{h_t(u,1)>0\}}, \qquad t \geq 0.$$

The gate is read in the update of $b_{t+1}(u, 0)$. Initialize the countdowns with the product law in Lemma 3.1, independently of all later marks, and initialize the bits with any deterministic configuration.

Lemma 3.2 (Exact gate law)**.** *The variables $G_t(u)$ are independent over $(u, t) \in \mathbb{Z}^2 \times \mathbb{N}$. They are independent of every replacement variable $R_s(v, 0), V_s(v, 0)$ in the bit plane. Their common closed-gate probability is*

$$q_H(\varepsilon) = \mathbb{P}(G_t(u) = 0) = \frac{1}{H + 1} \sum_{j=0}^{H} (1 - \varepsilon)^j = \frac{1 - (1 - \varepsilon)^{H+1}}{\varepsilon(H + 1)}. \tag{5}$$

From arbitrary deterministic initial countdowns, the same assertions hold for the gates with $t \geq H$.

Proof. The primitive variables determining $G_t(u)$ lie on the diagonal

$$(u, 1 + a, t - a), \qquad 0 \leq a < t,$$

together with the initial height at $(u, 1 + t)$ when it is needed. A second gate either has a different u or a different value of $z + t = 1 + t$. Its determining variables are then disjoint

from the first gate's. All initial countdowns are independent, so the gates are independent. Every determining site has $z \geq 1$, while the displayed bit-plane replacement variables have $z = 0$. This proves their independence as well.

By Lemma 3.1, a gate is closed exactly when $U \leq A$. Averaging first over U gives

$$\mathbb{P}(U \leq A) = \frac{1}{H+1} \sum_{j=0}^{H} \mathbb{P}(A \geq j) = \frac{1}{H+1} \sum_{j=0}^{H} r^j.$$

Summing this geometric series proves (5). For $t \geq H$, the gate depends only on the last H update positions, regardless of its initial countdown, so the preceding independence argument applies verbatim. □

The qualification about the initial countdowns is necessary. For example, from $h_0 \equiv H$, the gates at time zero are deterministically open. The stationary initialization gives the exact law (5) from the first bit update; arbitrary initialization gives it after the finite countdown transient.

Proposition 3.3 (The projected majority process)**.** *Under the stationary countdown initialization, $B_t(u) = b_t(u, 0)$ is a binary PCA with independent local random choices and update*

$$B_{t+1}(u) = \begin{cases} V_{t+1}(u, 0), & R_{t+1}(u, 0) = 1, \\ G_t(u)\,\mathrm{Maj}(B_t(u), B_t(u + e_1), B_t(u + e_2)), & R_{t+1}(u, 0) = 0. \end{cases} \tag{6}$$

The indicators

$$I_{t+1}(u) = \mathbf{1}_{\{R_{t+1}(u,0)=1 \text{ or } G_t(u)=0\}} \tag{7}$$

are independent Bernoulli variables of parameter

$$p_H(\varepsilon) = \varepsilon + (1 - \varepsilon) q_H(\varepsilon). \tag{8}$$

At every update where $I_{t+1}(u) = 0$, the bit obeys north-east-centre majority exactly.

Proof. Equation (6) is the bit coordinate of the original update. Its right-hand side uses the plane's bits, its past gate variables, and its plane replacement variables. By Lemma 3.2, the new gates and replacements are independent of all these earlier variables and independent across u. This proves the PCA assertion and (8). A zero indicator in (7) means an open gate and no replacement, giving the last assertion. □

For clarity, the independent indicators describe opportunities for an error; a replacement may happen to write the majority value. The exact transition probability of the projected PCA is

$$\mathbb{P}(B_{t+1}(u) = 1 \mid B_t) = \frac{\varepsilon}{2} + (1 - \varepsilon)(1 - q_H(\varepsilon))\,\mathrm{Maj}(B_t(u), B_t(u + e_1), B_t(u + e_2)). \tag{9}$$

The actual error probability is $\varepsilon/2$ when the majority is zero and $\varepsilon/2 + (1 - \varepsilon) q_H(\varepsilon)$ when it is one. Both are bounded by $p_H(\varepsilon)$.

4 Memory at a positive noise rate

We state precisely the stability estimate needed here, taken from Gács's exposition [1] of the Berman–Simon proof [5]. The parameter ρ below bounds local update errors relative to majority, and is distinct from the replacement probability ε of our model.

Theorem 4.1 (Toom stability, quantitative specialization)**.** *Let a binary process on $\mathbb{Z}^2$, started from the constant bit $a \in \{0,1\}$, be a ρ-perturbation of north-east-centre majority: conditional on the past, its site updates are independent and each disagrees with majority with probability at most ρ. If*

$$\rho < \tau := \frac{1}{32 \cdot 24^8}, \tag{10}$$

then at every site and time its probability of disagreeing with a is at most 96ρ.

This is the infinite-plane case $m = \infty$ of the main theorem of [1],[1] specialized as follows. Take the auxiliary one-dimensional space in its Definition 1.5 to be a singleton and its transition rule to be the identity. The resulting three-dimensional rule is exactly NEC majority on the remaining two coordinates. The constant reference trajectory is a at every site and time. The constants stated there are $r = 24$, threshold $(32r^8)^{-1}$, and error bound $4r\rho$. Independent site updates with the displayed conditional error bound satisfy its perturbation hypothesis, including conditioning on the other simultaneous outputs. This is the sole imported estimate; Toom's original theorem is [6]. A one-point auxiliary space is admitted by the definitions of [1]; the neighbours $n \pm 1$ then coincide with n, and its perturbation condition is read for the three-dimensional rule as in the theorem.

Theorem 4.2. *If $p_H(\varepsilon) < \tau$, then $P_{H,\varepsilon}$ has two translation-invariant stationary laws μ^0, μ^1 satisfying, at every site $v \in \mathbb{Z}^3$,*

$$\mu^0\{b(v) = 1\} \le 96 p_H(\varepsilon), \qquad \mu^1\{b(v) = 1\} \ge 1 - 96 p_H(\varepsilon). \tag{11}$$

In particular it is non-ergodic. The condition holds for

$$\varepsilon_* = 2^{-44}, \qquad H_* = 2^{88}. \tag{12}$$

Proof. For $a = 0, 1$, let ν^a be the initial law with every bit equal to a and countdowns drawn from the stationary product law. For every horizontal plane, Proposition 3.3 is a $p_H(\varepsilon)$-perturbation in the sense of Theorem 4.1. Its error probability conditional on the entire plane past is bounded by (8), and updates at the current time are conditionally independent. Thus

$$\mathbb{P}_{\nu^a}\{b_t(v) \ne a\} \le 96 p_H(\varepsilon) \qquad (v \in \mathbb{Z}^3,\ t \ge 0).$$

The planes need not be independent; the estimate uses their individual marginal processes.

Take a weak limit point μ^a of the Cesàro averages of $\nu^a P^t_{H,\varepsilon}$. The Feller argument in Section 2 proves stationarity. The initial laws are translation invariant and the transition commutes with translations, so the limit laws are also translation invariant. The one-site events in the preceding display are clopen cylinders, and their bounds pass to the limit, proving (11). Since $192\tau < 1$, the two bounds are disjoint and the laws differ.

For the explicit choice, (5) gives

$$q_{H_*}(\varepsilon_*) \le \frac{1}{\varepsilon_*(H_* + 1)} < 2^{-44}, \qquad p_{H_*}(\varepsilon_*) < 2^{-43}.$$

Also $32 \cdot 24^8 = 2^{29} 3^8 < 2^{42}$ because $3^8 = 6561 < 8192$. Hence $2^{-43} < \tau$, as required. □

More generally, for every fixed $0 < \varepsilon < \tau$, a sufficiently large H satisfies the hypothesis, since $q_H(\varepsilon) \le 1/[\varepsilon(H+1)]$. Thus the counterexample can be placed at arbitrarily small positive replacement rates by changing the automaton. For any one chosen H, sufficiently small noise has a different behavior, which we now prove. The existence of some H and ε with two stationary laws needs only Toom's theorem [6] with an unspecified threshold; the constants of [1] enter only in the explicit choice above.

[1] In the displayed perturbation condition of [1], the comparison with the prescribed output is printed as equality. We use the disagreement event specified by its subsequent definition of Noise and its proof.

5 Ergodicity at small and large noise

Couple two trajectories from arbitrary configurations by using the same replacement indicators and the same replacement symbols. Their countdowns agree everywhere by time H, even if no replacement occurs on a particular diagonal. Once the countdowns agree, a closed gate or a replacement makes the corresponding bit outputs agree. An open gate without a replacement can pass a disagreement only from one of the three majority parents in its own plane.

Put

$$\alpha_H(\varepsilon) = (1-\varepsilon)(1-q_H(\varepsilon)), \qquad \lambda_H(\varepsilon) = 3\alpha_H(\varepsilon). \tag{13}$$

The first quantity is the probability that one bit update can depend on its preceding bits.

Theorem 5.1. *If $\lambda_H(\varepsilon) < 1$, then $P_{H,\varepsilon}$ has a unique stationary law μ and, for every finite $W \subset \mathbb{Z}^3$ and $n \geq 1$,*

$$\sup_{x \in \Omega_H} d_{\mathrm{TV}}\big((\delta_x P_{H,\varepsilon}^{H+n})_W, \mu_W\big) \leq |W|\lambda_H(\varepsilon)^n. \tag{14}$$

It is uniformly exponentially ergodic whenever

$$0 < \varepsilon < \frac{2}{3H} \qquad \text{or} \qquad \frac{2}{3} < \varepsilon \leq 1. \tag{15}$$

The qualitative statements are known: ergodicity under small noise is [2, Theorem 3.9], and ergodicity under large noise holds for every perturbed cellular automaton. The explicit ranges and the coupling bound above are what we add.

Proof. Fix a site $v = (u, z)$ and suppose its bits in the two coupled trajectories differ at time $H + n$. Trace backward from this disagreement. At each step there must be an open gate, no replacement at the output, and at least one disagreeing majority parent. Consequently there is a path of length n ending at time H with each backward spatial displacement in $\{0, e_1, e_2\}$, all within the plane at height z. Every update on this path is open and unreplaced.

There are 3^n possible paths. For a fixed path, the relevant gates are those with input times $H, H+1, \ldots, H+n-1$ in one fixed plane. By the last assertion of Lemma 3.2, they are independent with open probability $1 - q_H(\varepsilon)$. They are independent of all replacements in that plane. The probability that every update on the path is open and unreplaced is therefore exactly $\alpha_H(\varepsilon)^n$. The union bound gives

$$\mathbb{P}\{b_{H+n}(v) \neq b'_{H+n}(v)\} \leq \lambda_H(\varepsilon)^n. \tag{16}$$

This argument does not require any independence between the initial disagreements and these gates: it bounds the event that a path of open, unreplaced updates exists. The countdowns already agree. Summing (16) over W bounds the probability of any full-symbol disagreement in W, uniformly over the two initial configurations.

A stationary law exists by compactness. Couple a deterministic initial configuration to a sample from that law and integrate the preceding uniform bound. The coupling inequality gives (14), hence convergence to this stationary law from every start. Coupling samples from any two stationary laws gives the same bound at every n; sending n to infinity proves uniqueness. If $0 < \lambda_H(\varepsilon) < 1$, take $c = -\log \lambda_H(\varepsilon)$ and enlarge the prefactor to cover $t \leq H$. If $\lambda_H(\varepsilon) = 0$, the trajectories agree after at most $H + 1$ steps, which also implies (3).

It remains to check (15). Bernoulli's inequality in (5) yields

$$q_H(\varepsilon) \geq \frac{1}{H+1} \sum_{j=0}^{H} (1 - j\varepsilon) = 1 - \frac{H\varepsilon}{2}.$$

Thus $\lambda_H(\varepsilon) \le 3H\varepsilon/2 < 1$ in the small-noise range. For large noise, $\lambda_H(\varepsilon) \le 3(1-\varepsilon) < 1$. □

Corollary 5.2 (Re-entrance)**.** *For the single rule F_{H_*} in (12), there is an open interval of positive noise rates containing ε_* on which the PCA is non-ergodic. It is uniformly exponentially ergodic at every rate in $(0, 2/(3H_*))$ and $(2/3, 1]$.*

Proof. For fixed H_* the expression $p_{H_*}(\varepsilon)$ is continuous on $(0, 1]$. The strict inequality proved in Theorem 4.2 persists in a neighborhood of ε_*. The two ergodic ranges follow from Theorem 5.1. They lie on either side of ε_* because $2/(3 \cdot 2^{88}) < 2^{-44} < 2/3$. □

These sufficient conditions establish ergodicity, non-ergodicity and ergodicity in that order as the noise rate increases. They do not locate the boundaries between the regimes. The exact contraction condition $3(1-\varepsilon)(1-q_H(\varepsilon)) < 1$ can also apply beyond the two intervals in (15).

6 Separated deterministic replacement events

A replacement event is a point $(u, z, s) \in \mathbb{Z}^2 \times \mathbb{Z} \times \mathbb{N}_{\ge 1}$ at which, after the deterministic update into time s, a prescribed symbol overwrites the output. All other updates follow F_H. Distances between events are measured in all four coordinates using ℓ^∞. In particular, separation in the z coordinate suffices even when events have the same time.

Theorem 6.1. *Given $d \ge 0$, choose an integer $m > d$ and let $H = m^4$. There is a set $\mathcal{E}$ of replacement events at pairwise ℓ^∞ distance at least m such that every event writes $(0, H)$, and the two initial configurations*

$$(b_0, h_0) \equiv (0, H), \qquad (b'_0, h'_0) \equiv (1, H)$$

produce $b_t(u, 0) = 0$ and $b'_t(u, 0) = 1$ for every $u \in \mathbb{Z}^2$ and $t \ge 0$ under these common events. The underlying rule is nilpotent of exponent $H + 1$.

Proof. For $u = (x, y)$ and $t \ge H$, let $[x]_m, [y]_m, [t]_m \in \{0, \dots, m-1\}$ be their residues, and set

$$c(u, t) = [x]_m + m[y]_m + m^2[t]_m, \qquad a(u, t) = m\, c(u, t). \tag{17}$$

The age satisfies $0 \le a(u, t) \le m^4 - m = H - m$. Place one event for each such pair (u, t) at

$$E(u, t) = \big(u, 1 + a(u, t), t - a(u, t)\big), \tag{18}$$

and let every event write $(0, H)$. Its time is at least $m \ge 1$, and its height coordinate is positive, so these are valid events and none belongs to the bit plane $z = 0$.

We first check separation. If two ages in (18) differ, their z coordinates differ by a nonzero multiple of m, hence by at least m. If the ages agree, the residue triples in (17) agree. Distinct pairs (u, t) then differ in at least one of x, y, t by a nonzero multiple of m. Equal ages preserve that difference in the corresponding event coordinates $x, y, s = t - a$. The events are therefore distinct and at pairwise distance at least $m > d$.

We next check that the gates into $z = 0$ are always open. With initial countdowns equal to H, and all subsequent resets equal to H, monotonicity of the decrement map gives

$$h_t(u, 1) \ge (H - t)_+.$$

This is positive for $0 \le t < H$. For $t \ge H$, the event $E(u, t)$ puts a countdown H on the backward diagonal of $(u, 1, t)$, at age $a = a(u, t) < H$. Propagating this value forward along that diagonal gives a countdown at least $H - a \ge m > 0$ at $(u, 1, t)$. Any intervening reset sets

the countdown to H and can only increase this lower bound. Hence every gate into the bit plane is open at every time.

The bits at $z = 0$ are never replaced and read bits only in that plane. They consequently follow exact NEC majority forever. The two constant initial bit planes remain respectively zero and one, proving the claimed discrepancy. Nilpotency of the rule without replacements is Proposition 2.1. □

The separation parameter is fixed before the automaton is chosen. Thus the theorem provides arbitrarily well separated renewal events across the family of nilpotent rules. Their positions in the extra spatial coordinate keep the renewals separated while feeding every gate of the same bit plane.

7 How this work was done

AI note. The authors chose the question, directed the work and judged every result. AI models from two developers assisted. OpenAI's GPT-6 proposed the countdown construction, and Anthropic's Claude Opus 5.5 checked it by hand. OpenAI's GPT-6.1 Sol, through Codex, searched the literature, wrote the proofs and prepared this manuscript. Anthropic's Claude Fable 5.2 read the manuscript refutation-first, in two rounds, before release, and its findings were fixed. Models are not listed as authors; the authors take responsibility for every claim. The proofs of the countdown reduction, coupling bound, and separated-event construction are given in full. The majority-stability estimate is imported from [1]; its specialization and constants are stated in Theorem 4.1.

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